INMO(7^{th}) – 1992

Time – 4 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. In a triangle ABC, angle A is twice the angle B. Show that $a^2 = b(b+c)$.
2. If x, y and z are three real numbers such that $x+y+z=4$ and $x^2+y^2+z^2=6$, then show that each of x, y and z lies in the closed interval [2/3, 2] that is $\frac{2}{3} \le x \le 2$, $\frac{2}{3} \le y \le 2$, $\frac{2}{3} \le z \le 2$. Can s attain the extreme value 2/3 or 2?
3. Find the remainder when 19^{92} is divided by 92.
4. Find the number of permutations $(P_1, P_2, P_3, P_4, P_5, P_6)$ of 1, 2, 3, 4, 5, 6 such that for any k, $1 \le k \le 5$. $(P_1, P_2, ..., P_k)$ does not form a permutation of {1, 2, 3, …., k}. That is $P_1 \ne 1$; (P_1, P_2) is not a permutation of {1, 2}; (P_1, P_2, P_3) is not a permutation of {1, 2, 3}.
5. Two circles C_1 and C_2 intersect at two distinct points P and Q in a plane. Let a line passing through P meet the circle C_1 and C_2 in A and B respectively. Let Y be the midpoint of AB and QY meets the circle C_1 and C_2 in X and Z respectively. Show that Y is also the midpoint of XZ.
6. Let f(x) be a polynomial in x with integer coefficients and suppose that for 5 distinct integers C_1 and a_1, a_2, a_3, a_4, a_5 one has $f(a_1) = f(a_2) = f(a_3) = f(a_4) = f(a_5) = 2$. Show that there does not exist an integer b such that f(b) =9.
7. Find the number of ways in which one can place the numbers $1,2,3,....,n^2$ on the n^2 squares of $n \times n$ chessboard, one on each, such that the numbers in each row and each column are in arithmetic progression. Assume $n \ge 3$.
8. Determine all pairs (m, n) of positive integers for which $2^m + 3^n$ is a perfect square.
9. Let $A_1A_2A_3...A_n$ be an n-sided regular polygon such that $\frac{1}{A_1A_2} = \frac{1}{A_1A_3} + \frac{1}{A_1A_4}$. Determine n, the number of the polynomial.
10. Determine all function $f : \Re \setminus \{0,1\} \to \Re$ satisfying the functional relation $f(x) + f\left(\frac{1}{1-x}\right) = \frac{2(1-2x)}{x(1-x)}$ where x is a real number different from 0 and 1 and $\Re$ is the set of real numbers.

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